

Comprehensive Road Scene Understanding for Autonomous Driving: From Mask-Based Segmentation to Anomaly Detection

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Abstract

Deep segmentation networks achieve excellent performance on the classes they were trained on, but behave unpredictably on unknown, out-of-distribution (OoD) objects a critical safety gap for autonomous driving, where an unexpected object on the road must be flagged rather than silently mis-labelled. In this project we (i) review semantic, instance and panoptic segmentation; (ii) compare two pre-trained *Encoder-only Mask Transformer* (EoMT) checkpoints one trained on Cityscapes, one on COCO under a single, consistent evaluation pipeline; (iii) fine-tune the COCO model toward the road domain; and (iv) build a suite of post-hoc anomaly-segmentation baselines on a pixel-based network (ERFNet) and on the mask-based EoMT, using four scoring rules (MSP, Max Logit, Max Entropy, Rejection-by-All) and temperature scaling, evaluated on five benchmarks with AuPRC and FPR₉₅.

We find that the mask architecture massively outperforms the pixel baseline (RoadAnomaly-21 AuPRC 38 → 77), that closed-set domain knowledge directly governs open-set detection (the off-domain COCO model is unusable until fine-tuned), and that the best scoring rule depends on how well calibrated the underlying classifier is. Source code: <https://github.com/kamyarfaz/anomaly-segmentation>.

1 Introduction

1.1 Project Context

Dense scene understanding underpins modern driver-assistance and self-driving systems. *Semantic segmentation* assigns a class label to every pixel, *instance segmentation* additionally separates distinct objects of the same class, and *panoptic segmentation* [11] unifies the two by labelling every pixel with both a class and (for “thing” classes) an instance identity. These tasks are largely solved in the closed world, where the set of classes fixed and known. Real driving, however, is an *open* world: a network trained on a fixed taxonomy will confidently misclassify any object outside it (animals, debris, lost cargo).

The goal of this project is therefore twofold to understand and compare modern mask-based segmentation models on known classes, and to design and evaluate an *anomaly-segmentation* pipeline that flags unknown regions in road scenes.

1.2 Literature Foundations

Real-time pixel-wise segmentation is represented here by ERFNet [10], a compact residual factorized CNN. A conceptual shift came with *mask classification*: MaskFormer [3] showed that predicting a set of binary masks, each with one class label, subsumes per-pixel classification, and Mask2Former [4] unified semantic, instance and panoptic segmentation through masked attention. Most recently, EoMT [6] demonstrated that a plain Vision Transformer initialised with the self-supervised features of DINOv2 [5] is “secretly” a segmentation model: object queries are appended directly to the ViT, removing the need for a specialised decoder. For anomaly detection we use *post-hoc* confidence scores that require no re-training: Maximum Softmax Probability (MSP) [14], the Max Logit score introduced when scaling OoD detection to real-world settings [8], predictive entropy, and — specific to mask models — Rejection-by-All (RbA) [7], which flags pixels that *no* known-class query claims. We evaluate on the SegmentMeIfYouCan [1] and Fishyscapes [2] benchmarks, with Cityscapes [12] and COCO [13] as in-distribution data, and consider parameter-efficient adaptation inspired by LoRA [9].

2 Methodology

2.1 Models

We study two architectures, **ERFNet** is an encoder-decoder CNN that outputs per-pixel class logits over the 19 Cityscapes training classes. **EoMT** couples a DINOv2 ViT-Base/14 backbone with N learnable queries; for each query it predicts a class distribution and a binary mask.

We use three EoMT checkpoints: trained on Cityscapes (semantic, $N=100$, 19 classes), trained on COCO (panoptic, $N=200$, 133 classes), and the fine-tuned model produced in Section 4.

2.2 From masks to pixels

To apply pixel-level scoring rules to EoMT we collapse its mask outputs into a per-pixel class score map

$$L_c(p) = \sum_{q=1}^N \sigma(m_q(p)) \text{softmax}(z_q)_c, \quad (1)$$

where m_q and z_q are the mask logits and class logits of query q and the “no-object” class is dropped. The per-pixel argmax of L gives the semantic prediction.

2.3 Post-hoc anomaly scores

Given per-pixel logits $\ell_c(p)$ (raw logits for ERFNet, L_c for EoMT) the anomaly score $s(p)$ is: *MSP* $s = 1 - \max_c \text{softmax}(\ell)_c$; *Max Logit* $s = -\max_c \ell_c$; *Max Entropy* $s = -\sum_c p_c \log p_c$ with $p = \text{softmax}(\ell)$; and, for mask models only, *RbA* $s = -\sum_c L_c(p)$ — a pixel is anomalous when the total response across all known classes is low. *Temperature scaling* replaces $\text{softmax}(\ell)$ with $\text{softmax}(\ell/T)$ before MSP.

2.4 Consistent cross-dataset evaluation

Our first implementation aimed to conduct a controlled comparison between two EoMT checkpoints trained on distinct datasets, Cityscapes and COCO, by evaluating them on identical Cityscapes validation images within an aligned label space. We initialized the architecture from its configuration, loaded the checkpoint weights, and generated per-pixel semantic predictions on resized inputs. To account for differing class vocabularies, we restricted the evaluation to a shared subset of known classes and mapped the COCO indices to Cityscapes identifiers, ensuring both models produced predictions in a common semantic space. This alignment enabled the calculation of a directly comparable Intersection over Union (IoU) metric for the shared classes, supplemented by qualitative visual comparisons of the predictions. Consequently, the pipeline successfully isolated the impact of the pretraining dataset—comparing in-domain against out-of-domain training—while keeping evaluation parameters constant for a rigorous analysis. A primary engineering challenge, common when adapting Vision Transformer (ViT) models, involved the tight coupling of checkpoints to input resolution. Adjusting the evaluation resolution alters the patch grid and the expected shape of the positional embeddings, causing loading failures; we resolved this by interpolating the checkpoint’s two-dimensional positional grid to match the target dimensions. Additionally, we encountered configuration inconsistencies, as

the two checkpoints required precise alignment of meta-data and explicit initialization parameters rather than default settings. Reconciling these label-set discrepancies and varying output dimensions was necessary to make cross-dataset metric comparisons meaningful. In contrast, the subsequent notebook-based approach presents distinct practical advantages. It loads the network directly with minimal overhead and processes inputs at the model’s native square resolution, thereby mitigating positional embedding conflicts. Furthermore, it efficiently constructs a unified per-pixel score map from a single forward pass, which can seamlessly inform multiple anomaly scoring rules, making it highly suitable for anomaly segmentation evaluation pipelines.

The other approach frames the Cityscapes-vs-COCO question as a closed-set semantic generalization problem, evaluated directly on Cityscapes validation images. Two EoMT checkpoints are instantiated from their respective configurations—a Cityscapes-trained semantic model and a COCO-trained panoptic model—and both are forced into a common evaluation space by projecting predictions onto Cityscapes trainIds. Because the COCO model outputs a larger label set, its predicted indices are mapped into the subset of Cityscapes classes that have clear correspondences, yielding a 16-class shared label space (with classes lacking a reliable COCO equivalent excluded). Table 4 in appendix for more information.

2.5 Fine-tuning

The approach was to implement a constrained fine-tuning strategy intended to reduce the domain and label-space gap between the COCO-trained EoMT checkpoint and Cityscapes, while keeping the optimization problem tractable on limited compute. The procedure reloads the COCO model to preserve its learned mask queries and backbone representations, then freezes the full network and replaces only the classification head with a new linear layer sized for Cityscapes, namely 19 classes plus the no-object channel. This design deliberately limits the number of trainable parameters, on the order of 10^4 , to avoid the instability and memory cost that would arise from updating the transformer backbone and query/mask blocks. Therefore, the adaptation is treated primarily as a recalibration of class semantics rather than a full representational shift.

Training data are drawn from the Cityscapes `train` split using a fixed-size subset for speed. Both images and labels are processed through the same letterboxing-to-square scheme used at inference to avoid aspect-ratio distortion. Labels are remapped from Cityscapes `labelIds` to `trainIds`, padded consistently with the image letterbox, and downsampled to a lower-resolution supervision grid to match the computational budget of query-mask decoding.

The optimization objective is defined on a dense se-

mantic score map derived from the query-based outputs. The model’s final-layer masks and class logits are combined into per-pixel class probabilities by weighting each query mask with its class distribution. The output is then restricted to the first 19 semantic channels and renormalized to form a proper distribution over Cityscapes classes. A per-pixel negative log-likelihood, equivalently cross-entropy on log-probabilities, is computed while ignoring padded and void pixels. Mixed-precision training is used to accelerate updates and reduce GPU memory footprint.

Conceptually, this makes the head learn how to reinterpret COCO-derived query embeddings as Cityscapes categories under the same mask decomposition. This is a pragmatic compromise: it can improve closed-set semantic alignment without requiring the model to relearn instance partitioning or mask generation from scarce supervision.

The same constraints also expose the key fine-tuning difficulties and limitations. Because only the classification head is updated, the approach cannot correct backbone-level domain shift, including appearance statistics, scene layout, and long-tail category cues. It also cannot correct query/mask allocation behavior that may be suboptimal for Cityscapes. Consequently, gains are expected to be bounded, particularly for classes whose representation depends on geometry and context learned differently under COCO.

The loss is applied to a downsampled target grid and uses a simplified semantic collapse of a panoptic-style model. Therefore, improvements in mIoU do not necessarily translate into better instance separation or anomaly scoring. The quality of adaptation depends strongly on the correctness of the label remapping, the handling of no-object mass, and the extent to which letterboxing-induced padding is excluded from supervision.

In practice, this head-only fine-tuning is attractive because it is fast, reproducible, and comparatively stable. However, it remains a partial solution, and the remaining performance gap reflects the fundamental mismatch between pretraining supervision, label space, and the target driving-scene distribution.

2.6 Metrics

Closed-set quality uses *mean Intersection-over-Union* (mIoU): the average, over classes, of (correct pixels)/(predicted+true−correct). Anomaly quality uses *AuPRC* (area under the precision–recall curve; higher is better, robust to the strong class imbalance of anomaly pixels) and *FPR₉₅* (false-positive rate at 95% true-positive rate; lower is better, i.e. how many normal pixels are wrongly flagged once most anomalies are caught).

Table 1: Per-class IoU and mIoU (%) on the 16 shared classes of Cityscapes val. FT = COCO model after head fine-tuning.

Class	City-EoMT	COCO-EoMT	COCO→FT
road	98.1	93.4	96.3
sidewalk	84.8	61.3	72.3
building	92.2	83.1	86.1
wall	63.8	47.9	36.0
fence	61.8	39.2	44.5
traf. light	65.9	45.8	47.2
vegetation	91.0	84.3	85.5
terrain	62.8	53.6	56.7
sky	93.9	87.0	89.4
person	78.1	60.0	65.3
car	93.8	80.7	88.4
truck	78.5	27.3	38.7
bus	87.8	64.8	81.5
train	77.8	4.2	68.1
motorcycle	64.4	51.7	52.9
bicycle	74.3	61.9	62.2
mIoU (16)	79.3	59.2	66.9

3 Experimental Results

3.1 Results reporting

Table 1 reports the closed-set mIoU comparison (Steps 4–5). Table 2 reports the full anomaly baselines for ERFNet and the three EoMT checkpoints (Steps 7–8). Table 3 reports temperature scaling. Figure 1 shows a qualitative anomaly example, Figure 2 the qualitative panoptic comparison of the two checkpoints, and Figure 3 a per-model anomaly-heatmap comparison. As a sanity check, ERFNet reaches 72.2 mIoU over all 19 classes, matching its published score and validating the pipeline.

3.2 Quantitative Breakdown

On the closed-set task, the Cityscapes model leads with 79.3 mIoU, the COCO model trails at 59.2, and fine-tuning recovers a large part of the gap to 66.9. The COCO model’s weakness is concentrated on road-specific and dynamic classes (**truck** 27, **train** 4, **person** 60), and fine-tuning improves exactly those (**train** 4 → 68, **bus** 65 → 82, **person** 60 → 65), while leaving classes COCO already knew nearly unchanged. On anomaly detection, the mask-based EoMT dominates the pixel-based ERFNet on every benchmark (RA-21 AuPRC 77 vs. 38; Road Anomaly 78 vs. 16). The COCO model is essentially unusable on road anomalies (RO-21 AuPRC 1.4, FPR₉₅ ≈ 100), whereas fine-tuning lifts it substantially (RO-21 1.4 → 56, RA-21 34 → 44, Road Anomaly 21 → 38). Temperature scaling changes AuPRC by less than 0.3 points.

Table 2: Anomaly segmentation: AuPRC ($\uparrow$) / FPR₉₅ ($\downarrow$), in %. mIoU is the closed-set quality of each model (constant across scoring rules). Best AuPRC per benchmark, within each model block, in **bold**.

Model	mIoU	Method	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR	AuPRC	FPR	AuPRC	FPR	AuPRC	FPR	AuPRC	FPR
ERFNet	72.2	MSP	29.08	62.57	2.71	65.30	1.73	50.60	5.03	40.93	12.42	82.59
		Max Logit	38.29	59.37	4.63	48.47	3.28	45.47	6.27	39.68	15.58	73.28
		Max Entropy	30.95	62.68	3.05	65.96	2.56	50.17	5.89	40.65	12.67	82.76
EoMT-City	79.3	MSP	74.25	33.45	90.18	0.53	24.64	15.36	29.90	34.50	75.76	19.36
		Max Logit	73.71	33.57	90.14	0.56	24.66	15.10	30.12	34.17	75.13	19.11
		Max Entropy	76.69	33.33	90.10	0.68	27.48	15.18	28.65	34.42	77.66	18.85
		RbA	72.82	45.67	88.63	69.37	25.12	11.24	32.60	34.77	75.03	17.30
EoMT-COCO	59.2	MSP	34.35	87.96	1.43	99.99	1.42	96.73	4.13	97.11	20.73	94.03
		Max Logit	34.95	87.78	1.42	99.99	1.41	96.74	4.10	97.19	20.99	94.09
		Max Entropy	41.00	80.13	3.72	99.99	1.14	96.51	3.88	97.94	23.38	92.69
		RbA	37.43	62.23	36.76	100.0	0.28	96.65	1.27	99.95	20.65	84.39
EoMT-FT	66.9	MSP	44.08	31.81	55.83	99.98	0.31	87.97	29.15	54.81	38.16	90.86
		Max Logit	41.26	33.19	51.11	99.99	0.29	89.00	25.66	55.04	35.35	92.07
		Max Entropy	38.79	34.05	49.50	99.99	0.29	88.56	22.66	54.82	34.15	92.77
		RbA	16.01	88.99	2.90	100.0	0.19	98.80	4.31	79.69	18.03	97.31

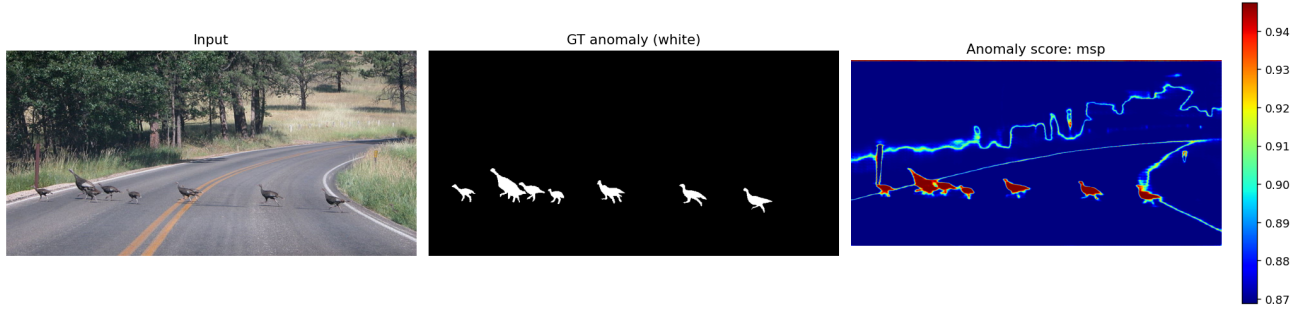


Figure 1: Qualitative anomaly segmentation (EoMT-Cityscapes, MSP) on RoadAnomaly-21: input, ground-truth anomaly mask, and the predicted anomaly heatmap. The unknown objects (animals) are correctly highlighted, with some false positives on far-away background structures.

Table 3: Temperature scaling of MSP (EoMT-Cityscapes), AuPRC (%).

T	RA-21	RO-21	FS L&F	FS Static	R.Anom
0.50	74.14	90.18	24.62	29.87	75.60
0.75	74.20	90.18	24.63	29.89	75.69
1.00	74.25	90.18	24.64	29.90	75.76
1.10	74.27	90.18	24.64	29.91	75.78
2.00	74.38	90.18	24.66	29.93	75.91

4 Discussion

4.1 Result Analysis

Two patterns dominate. First, the mask architecture wins because of *what* and *how* it represents. Its DINOv2 backbone provides rich self-supervised features that generalise beyond the training labels, so unknown objects produce distinctive responses; and mask classification decouples “where” (masks) from “what” (classes), so an unknown region can simply be claimed by no class — exactly the signal RbA exploits. ERFNet, forced to assign every pixel to one of 19 classes, produces over-confident predictions with little usable uncertainty. Second, and most im-

portantly, closed-set knowledge *drives* open-set detection: the COCO model never learned what a normal road looks like, so it cannot recognise what deviates from it, and its anomaly scores are near-random. After fine-tuning only a 15k-parameter head, both its mIoU and its anomaly detection improve markedly, confirming that OoD detection is fundamentally relative to a well-modelled in-distribution.

The scoring-rule comparison adds nuance. Max Entropy and MSP are the strongest rules on the well-trained Cityscapes model, while RbA, although competitive there, *collapses* on the fine-tuned model (RA-21 AuPRC 16 vs. MSP’s 44). RbA sums un-normalised class responses and therefore depends on the classifier’s calibration, which our lightweight head training did not preserve; rules that read the maximum or the shape of the distribution are far more robust. Figure 3 makes this visible: the fine-tuned model flags anomalies strongly (high AuPRC) but treats much of the image as suspicious (high FPR₉₅). The flatness of temperature scaling shares this root cause — L is already a product of sigmoid and softmax terms, leaving little for a second temperature to reshape.

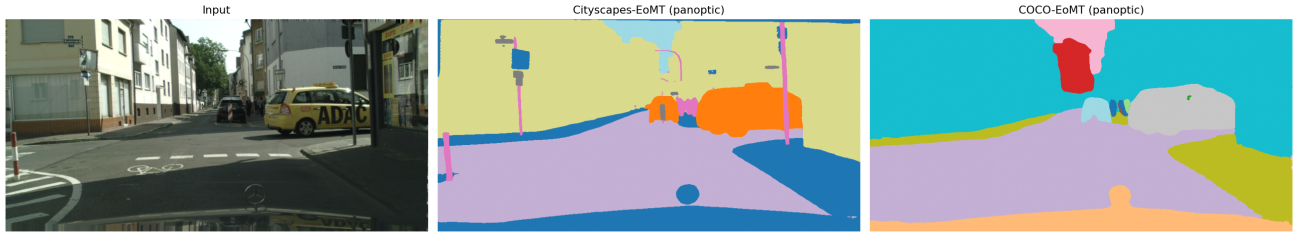


Figure 2: Qualitative panoptic comparison on a Cityscapes image. Each pixel is coloured by the query that owns it. The two checkpoints partition the scene differently: the Cityscapes (semantic) model yields coherent road-scene regions, while the COCO (panoptic) model groups the scene according to its own 200-query, 133-class taxonomy. Colours are query identifiers and are not comparable across panels.

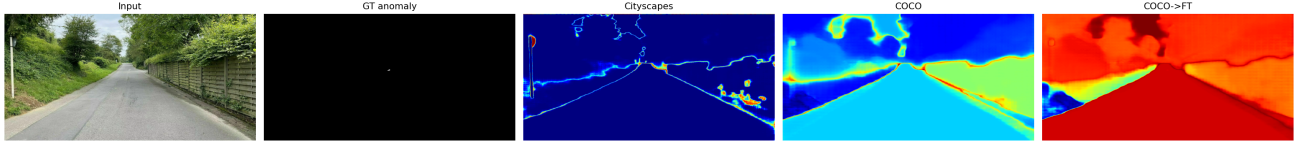


Figure 3: Per-model anomaly heatmaps (MSP) on a RoadObstacle-21 image. The Cityscapes model keeps the road “cold” (low anomaly); the COCO model fails to separate normal from abnormal; the fine-tuned model reacts strongly but is globally over-confident in flagging anomalies, explaining its high AuPRC yet poor FPR₉₅.

4.2 Methodological Critique

Several simplifications bound our numbers. (i) We use a single letterboxed forward pass instead of EoMT’s sliding-window inference, so absolute mIoU is below the official figures — but the protocol is identical for all models, preserving fairness. (ii) Fine-tuning trains only the classification head with a per-pixel cross-entropy (not the full mask-matching objective) on a 1000-image subset; the frozen COCO masks cap the achievable mIoU at 66.9, and the head’s random initialisation makes the fine-tuned numbers vary slightly between runs. (iii) Our RbA uses the summed-softmax variant; the original multi-label formulation may behave differently. (iv) FS Lost&Found stays hard for all COCO-derived models because its anomalies are small and distant. (v) FPR₉₅ is often near 100% even when AuPRC is good, showing that the score *ranking* is informative while the operating *threshold* is poorly calibrated — a clear direction for future work (fuller fine-tuning, e.g. LoRA, and explicit calibration).

5 Conclusion

We built and evaluated a complete road-scene anomaly-segmentation pipeline, comparing a pixel-based network with a state-of-the-art mask transformer across four scoring rules and three checkpoints. Mask-based EoMT dramatically outperforms ERFNet, closed-set domain knowledge is the key driver of open-set detection, and the best scoring rule depends on model calibration. A lightweight head-only fine-tuning recovers most of the COCO model’s domain gap at negligible cost, while underlining that reliable anomaly detection requires calibration, not accuracy

alone.

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- generating or refactoring LaTeX code snippets without changing the technical meaning.
- Tool(s) used:** Gemini (model/version: pro 3.1).

Acknowledgement

During this project, AI tools were used in a limited, support-only capacity. Specifically, they were used to troubleshoot implementation issues, such as configuration and checkpoint-loading problems, accelerate minor refactoring and code organization, improve the clarity and correctness of written material (grammar, spelling, and light rephrasing), and assist with LaTeX formatting.

Approach 1 – Task 4: Initial Script-Based Cityscapes-vs-COCO Comparison

AI assistance was used to address engineering and debugging challenges encountered while implementing the evaluation pipeline. Examples include resolving configuration-related issues, such as YAML/config structure mismatches, and suggesting fixes for model/checkpoint loading errors, such as tensor shape incompatibilities during state-dict loading.

Tool(s) used: ChatGPT (GPT-5.2).

Approach 2 – Chosen Notebook Approach: Step 4 Closed-Set Evaluation and Step 5 Fine-Tuning

AI assistance was used to support the transition from the initial approach to the notebook-based approach and to help interpret and describe the notebook implementation in academic writing. Concretely, AI was used to summarize the logic of the closed-set evaluation, including class mapping, letterboxing, and mIoU computation, and the fine-tuning setup, including the frozen model, head replacement, and loss definition. AI was also used to help identify potential limitations to report, such as dependence on class mappings, resolution differences, and constraints of head-only fine-tuning.

Tool(s) used: ChatGPT (model/version: plus 5.5); and/or Claude (model/version: opus 4.7 (High)); and/or Gemini (model/version: pro 3.1).

Writing, Editing, and Formatting

This applies to both approaches. AI tools were used only for surface-level writing support, including fixing grammar and spelling, minor rephrasing for readability, and